

# Financial Machine Learning: Data Structures, Advanced Labeling & Statistical Foundations

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## Chapter 2 Overview

### The Data Pipeline Problem

Raw financial data (tick-by-tick transactions) is highly irregular, noisy, and non-IID. We must transform this unstructured flow into structured data bars optimal for machine learning algorithms.

### Structure of this Section:

- **Textbook Background:** Volatility Clustering, Microstructure & Corporate Actions
- **Standard Bars:** Time, Tick, Volume, and Dollar Bars
- **Information-Driven Bars:** Imbalance and Runs Bars
- **Multi-Product Series:** The ETF Trick
- **Event-Based Sampling:** The CUSUM Filter
- **Simulation Demo:** Quantifying Normality Recovery

## Ch.2 Background 1: Volatility Clustering & Time Bars

### Academic vs. Empirical Time

Standard econometrics samples data at equal intervals of chronological time (e.g., daily, hourly). However, markets do not process information at a constant chronological pace.

- **Volatility Clustering (Heteroscedasticity):** Financial markets exhibit structural variations in activity. High-volatility periods cluster together, followed by quiet periods.
- **Intraday Seasonality (The U-Shape):** Volume and volatility spike near market Open and Close, while drying up during mid-day.
- **The Consequence:** Sampling by constant chronological intervals forces us to over-sample quiet periods (adding noise) and under-sample volatile periods (missing key information). This violates the IID normal distribution assumption required by ML.

## Ch.2 Background 2: Market Microstructure & Order Book

- **The Limit Order Book (LOB):**

- **Bids** represent buyers' resting limit orders; **Asks** represent sellers' resting limit orders.
- **Bid-Ask Spread:** The friction cost of immediate execution. Crossing the spread (buying at Ask, selling at Bid) represents an immediate transaction cost.

- **Order Fragmentation:**

- Institutional algorithms split massive trades into thousands of micro-orders (ticks) to hide their market impact.
- This means **Tick counts are heavily manipulated** by trading systems and do not accurately represent the true economic volume exchanged.

## Ch.2 Background 3: Corporate Actions & Price Adjustments

- **Corporate Actions:**

- **Stock Splits / Reverse Splits:** Artificially change nominal stock prices and outstanding shares without altering company equity.
- **Dividends:** Cash distributions that instantly drop the stock price on the ex-dividend date.

- **Data Adjustment and Its Pitfall:**

- Back-adjusting historical prices is necessary but can introduce **Look-ahead Bias** if future adjustment factors are accidentally leaked into historical trading simulations.
- Standard Volume and Tick bars are vulnerable to sudden jumps during splits, whereas **Dollar Bars** (tracking total currency exchanged) adjust dynamically and remain stable.

## Topic 2.1: The Four Essential Data Types

To feed our ML models, we must understand the taxonomy of financial data:

| Data Type   | Description                                     | ML Challenges                          |
|-------------|-------------------------------------------------|----------------------------------------|
| Fundamental | Balance sheets, earnings, macro indicators      | Low frequency, revisions, lag          |
| Market      | L3/L2 Order book feed, executed ticks           | High frequency, noise, size            |
| Analytics   | Sentiment scores, news signals, analyst ratings | Highly subjective, sparse              |
| Alternative | Satellite imagery, credit cards, shipping data  | High cost, unstructured, short history |

## Topic 2.2: Standard Data Bars (Why Use Dollar Bars?)

Standard bars aggregate tick-by-tick transactions based on a fixed threshold:

- **Time Bars:** Sampled every  $t$  minutes. Suffer from heavy non-normality.
- **Tick Bars:** Sampled every  $N$  transactions. Captures transaction counts, but distorted by order splitting.
- **Volume Bars:** Sampled every  $V$  shares/contracts traded. Captures size, but ignores nominal value.
- **Dollar Bars (The de Prado Recommendation):**
  - Sampled every  $D$  currency units (e.g., \$10M) exchanged.
  - **Self-Adjusting:** As price inflation or splits occur, the share count automatically adjusts, maintaining a stable sampling rate.
  - **Statistical Superiority:** Dollar-bar returns exhibit distributions much closer to a Gaussian normal distribution with stable variance (homoscedasticity).

## Topic 2.3: Information-Driven Bars (Imbalance Concept)

### Tick Rule & Order Flow Imbalance

Let  $b_t \in \{-1, +1\}$  be the trade sign via the tick rule:

$$b_t = \begin{cases} b_{t-1} & \text{if } \Delta p_t = 0 \\ \frac{|\Delta p_t|}{\Delta p_t} & \text{if } \Delta p_t \neq 0 \end{cases}$$

The cumulative tick imbalance over a bar of length  $T$  is  $H_T = \sum_{t=1}^T b_t$ .

### Tick Imbalance Bar (TIB) Condition

Sample a bar when the absolute cumulative imbalance reaches its expected threshold:

$$|H_T| \geq E_0[T] \cdot |2P[b_t = 1] - 1|$$

(For Volume/Dollar Imbalance Bars, weight  $b_t$  by volume  $v_t$  or dollar value  $v_t p_t$ ).

## Topic 2.3: Information-Driven Bars (Runs Concept)

While imbalance bars look at net signed flow, **Runs Bars** measure the sequence of consecutive buys or sells to capture institutional order-splitting algorithms.

### Sequence of Runs Definition

$$\text{Let } \theta_T = \max \left( \sum_{t \in \{t | b_t = 1\}} b_t, - \sum_{t \in \{t | b_t = -1\}} b_t \right)$$

### Tick Runs Bar (TRB) Condition

We sample a bar when the maximum run sequence exceeds expected parameters:

$$\theta_T \geq E_0[T] \cdot \max(P[b_t = 1], 1 - P[b_t = 1])$$

(For Volume/Dollar Runs Bars, weight by  $v_t$  or  $v_t p_t$  with expected directional size  $E_0[v_t | b_t = \pm 1]$ ).

## Topic 2.4: Multi-Product Series (The ETF Trick)

### The Practical Challenge

ML models cannot easily predict multiple raw price series that have differing contract sizes, transaction costs, rollovers, and weightings (e.g., spread trading or physical portfolios).

- **The ETF Trick:**

- A rigorous mathematical framework that models a complex, dynamic basket of assets (with rebalancing, dividend reinvestment, and funding costs) as a single, continuous, investable 1D price series.
- Incorporates transaction costs (crossing the bid-ask spread) at each rebalancing.
- This creates a clean, tradeable representation of a trading strategy's equity curve, which can then be predicted by ML models without look-ahead bias.

## Topic 2.5: Event-Based Sampling (The CUSUM Filter)

To avoid over-fitting and reduce computational cost, we should only feed the ML model data points that represent a genuine structural event or price breakthrough.

### Symmetric Cumulative Sum (CUSUM) Filter

An event-detection scheme that identifies local regime shifts in a log-price series  $y_t = \ln p_t$ :

$$S_t^+ = \max(0, S_{t-1}^+ + y_t - y_{t-1} - e)$$

$$S_t^- = \min(0, S_{t-1}^- + y_t - y_{t-1} + e)$$

An event is sampled when  $\max(S_t^+, -S_t^-) \geq h$ , where  $h$  is a threshold (often scaled by rolling volatility  $\sigma_t$ ) and  $e$  is a drift tolerance parameter.

## Empirical Demonstration: Time Bars vs. Dollar Bars

### Statistical Summary of Normality Recovery

By aligning the sampling clock with transaction value flow rather than physical time, Dollar Bars suppress intraday heteroscedasticity, producing near-Gaussian returns.

| Sampling Method         | Skewness | Excess Kurtosis | Jarque-Bera p-val | Normality       |
|-------------------------|----------|-----------------|-------------------|-----------------|
| Time Bars (10-Min)      | -0.12    | 1.85            | $< 0.0001$        | Rejected        |
| Dollar Bars (Matched N) | +0.02    | 0.20            | 0.8473            | <b>Accepted</b> |

**Note:** Realized excess kurtosis is cut by approximately **89%** using Dollar Bars, resolving tail risks for ML models.

# Python Code: Simulating Normality Recovery

```
import numpy as np
import pandas as pd
from scipy import stats

# Calculate Log Returns
time_returns = np.diff(np.log(time_bars['price']))
dollar_returns = np.diff(np.log(dollar_bars['price']))

# Test Normality via Jarque-Bera
time_jb = stats.jarque_bera(time_returns)
dollar_jb = stats.jarque_bera(dollar_returns)

print(f"Time Bars -> Kurtosis: {stats.kurtosis(time_returns):.2f}, JB p-val: {time_jb.pvalue:.4f}")
print(f"Dollar Bars -> Kurtosis: {stats.kurtosis(dollar_returns):.2f}, JB p-val: {dollar_jb.pvalue:.4f}")
```

## Chapter 3 Overview

### The Target Problem

A machine learning model cannot predict "prices." It must predict structured labels ( $y$ ) representing investment outcomes. How we define these labels determines whether our model is tradeable or merely a mathematical illusion.

### Structure of this Section:

- **Textbook Background:** Stop-Loss/Profit-Taking, Path-Dependency & Bet Sizing
- **Traditional Labeling:** The Fixed-Time Horizon Method and its Flaws
- **Dynamic Thresholds:** Volatility-adjusted Barriers
- **The Triple-Barrier Method:** Modeling Real Trading Rules
- **Meta-Labeling:** Decoupling Trade Direction (Side) from Bet Size
- **Class Imbalance Mitigation:** Dropping Unnecessary Labels

## Ch.3 Background 1: Trading Rules & Path-Dependency

### European vs. Barrier Options Analogy

Traditional finance research treats investment labels like European options (evaluating outcomes only at a fixed end-point). Real-world trading, however, behaves like Barrier options.

- **Profit-Taking & Stop-Loss Rules:** Every institutional trade is bound by strict risk management boundaries. If a position drops too far, it is immediately closed (Stop-loss). If it hits the target, profits are realized (Profit-taking).
- **Path-Dependency:** The path of the price matters. Even if a stock ends up 20% higher in 30 days, if it dipped 50% on day 3, a real-world trader would have been liquidated. We must use path-dependent labeling to account for this.

## Ch.3 Background 2: Bet Sizing & The Kelly Criterion

### Decoupling Direction from Bet Sizing

Finding a profitable opportunity (Is the market going up or down?) is only half the battle. Surviving the market requires deciding how much to bet (Bet Sizing).

- **The Danger of Uniform Betting:** Betting the same amount on every signal regardless of model confidence leads to rapid bankruptcy during a string of bad luck.
- **The Kelly Criterion:** A classical formula that dictates the optimal bet size ( $f^*$ ) based on the winning probability ( $p$ ) and the win-to-loss ratio ( $b$ ):

$$f^* = \frac{p \cdot b - (1 - p)}{b} = \frac{p(b + 1) - 1}{b}$$

- **The Role of ML:** We need our labeling system to feed the ML model not just the direction, but the probability of success to feed into our bet sizing algorithm.

## Ch.3 Background 3: ML Metrics vs. Investment Metrics

### The Misalignment

A machine learning engineer optimizes for Accuracy. A portfolio manager optimizes for the Sharpe Ratio and minimum drawdown. These are not always aligned.

- **The Fallacy of High Accuracy:** A model with 90% accuracy can still bankrupt a fund if its 10% failures are catastrophic drawdowns.
- **Precision vs. Recall in Finance:**
  - **Recall (Opportunity capture):** "Out of all profitable trades, how many did we take?"
  - **Precision (Quality of bets):** "Out of all trades we took, how many were actually profitable?"
- **F1-Score to Sharpe Ratio:** Maximizing the F1-Score (balance of Precision & Recall) directly translates to a higher Sharpe Ratio by reducing transaction costs and eliminating false positives.

## Topic 3.1: The Fixed-Time Horizon Method & Its Flaws

### The Traditional Academic Approach

$$y_i = \begin{cases} 1 & \text{if } P_{i+h} > P_i + \tau \\ -1 & \text{if } P_{i+h} < P_i - \tau \\ 0 & \text{otherwise} \end{cases} \quad (\text{where } \tau \text{ is a constant threshold}).$$

- **Constant Threshold Illusion:** A fixed threshold ( $\tau = 1\%$ ) in a low-volatility regime will produce only 0 labels. In a high-volatility regime, it will produce random  $\pm 1$  labels, breaking stationarity.
- **No Intra-horizon Stops:** It assumes the portfolio manager holds blindly until time  $h$ , completely ignoring intermediate stop-loss or profit-taking triggers.

### The Solution

Make profit/loss thresholds dynamic (based on rolling volatility) and labeling path-dependent (tracking intra-horizon price action).

## Topic 3.2 & 3.3: The Triple-Barrier Method

To model realistic trading, de Prado introduces the Triple-Barrier Method with three barriers:

- ① **Upper Horizontal Barrier:** Profit-taking limit ( $\pi_+ = p_0 \cdot [1 + \theta_1 \sigma_t]$ )
- ② **Lower Horizontal Barrier:** Stop-loss limit ( $\pi_- = p_0 \cdot [1 - \theta_2 \sigma_t]$ )
- ③ **Vertical Barrier:** Maximum holding time / expiration ( $t_1 = t_0 + H$ )

### Label Determination

$$y_i = \begin{cases} +1 & \text{if Upper Barrier touched first (Profit-Taking)} \\ -1 & \text{if Lower Barrier touched first (Stop-Loss)} \\ 0 \text{ or } \text{sgn}(r_{t_0, t_1}) & \text{if Vertical Barrier touched first (Expiration)} \end{cases}$$

**Dynamic Volatility Adjustment:**  $\sigma_t$  is the EWMA volatility of returns, scaling barriers dynamically to adapt to changing market volatility regimes.

## Topic 3.4: Meta-Labeling (Decoupling Side & Size)

### Two-Stage Learning Architecture

Instead of asking one ML model to determine both direction and bet size, we split the problem into a primary and secondary model.

#### ① Primary Model (Primary Signal):

- Technical rule (e.g., SMA crossover) or fundamental signal.
- Determines the **Side** ( $s_i \in \{-1, +1\}$ : Buy or Sell).

#### ② Secondary Model (Meta-Labeling):

- Apply Triple-Barrier Method **only in the direction** of the primary signal.
- If trade hits profit-taking:  $y_i^{\text{meta}} = 1$  (True Positive).
- If trade hits stop-loss or expires with loss:  $y_i^{\text{meta}} = 0$  (False Positive).
- Secondary ML model learns  $P[y_i^{\text{meta}} = 1]$  to optimize **Bet Size** / execution.

# The Superpowers of Meta-Labeling

Why is Meta-Labeling the preferred architecture of quantitative hedge funds?

- **Mitigates Class Imbalance:** The secondary model only evaluates trades when a primary signal occurs, focusing purely on filtering bad trades (optimizing precision).
- **Enables "Quantamental" Trading:**
  - Keeps the primary model as a clear, intuitive, economically-justified "white box" (e.g., MACD or fundamental value).
  - Overlays a sophisticated ML "black-box" as a risk filter (secondary model) to size the bet.
- **Prevents Overfitting:** Since the ML model does not decide the direction (which is noisy), but only whether to execute and how much to bet, model variance and overfitting risks are minimized.

# Graduate Advanced Overview

## The Deep Statistical Reality

To achieve mathematical rigor at the PhD/Graduate level, we must resolve fundamental violations of classical probability theory in financial machine learning.

- ❶ **The Stationarity vs. Memory Dilemma:** Standard price series are non-stationary, but standard return series (differencing  $d = 1$ ) erase all long-term historical memory.
- ❷ **Non-IID Sample Overlap:** Overlapping labels in the Triple-Barrier Method lead to highly correlated training observations, causing Bagging (Random Forest) failure.
- ❸ **Empirical Covariance Instability:** Raw empirical covariance matrices are dominated by noise, destroying Markowitz portfolio optimization.

## Topic 4.1 & 4.2: Fractional Differentiation (FFD)

To make  $X_t$  stationary, standard practice is integer differencing  $d = 1$  (returns). However, this wipes out long-term historical memory needed for predictive modeling.

### Fractional Differentiation ( $0 < d < 1$ )

$$(1 - B)^d X_t = \sum_{k=0}^{\infty} \omega_k X_{t-k} \quad \text{where } \omega_0 = 1, \quad \omega_k = -\omega_{k-1} \frac{d - k + 1}{k}$$

### Fixed-Width Window FFD ( $\tilde{X}_t$ )

$$\tilde{X}_t = \sum_{k=0}^l \omega_k X_{t-k} \quad (\text{dropping terms where } |\omega_k| < \tau, \text{ e.g., } 10^{-4}).$$

**The Benefit:** FFD finds the exact minimum  $d^*$  that achieves stationarity (ADF test passed) while preserving over 90% of the original price's correlation and long-term memory.

## Topic 4.3: Sequential Bootstrap (Label Uniqueness)

- **The Concurrency Problem:** If label 1 spans  $t_1 \rightarrow t_2$ , and label 2 spans  $(t_1 + 1) \rightarrow (t_2 + 1)$ , they share underlying market returns and are not independent.
- **Failure of Standard Bagging:** Uniform bootstrap sampling with replacement assumes IID data. Under overlapping labels, it draws highly redundant samples, leading to massive out-of-sample over-fitting.

### The Sequential Bootstrap Solution

Instead of uniform probability, dynamically calculate the **Uniqueness**  $u_i^{(t)}$  of sample  $i$  relative to the already drawn sample set:

- 1 Compute the concurrency of active labels at each timestamp.
- 2 Calculate the uniqueness  $u_i^{(t)}$  of each observation.
- 3 Sample the next row with probability proportional to its uniqueness.

## Topic 4.4: Random Matrix Theory (Denoising Covariances)

Empirical covariance matrices  $\Sigma$  from asset returns are highly unstable and noisy.

### The Marcenko-Pastur Theorem

For a random matrix with dimensions  $T \times N$ , as  $T, N \rightarrow \infty$  with  $q = \frac{T}{N} > 1$ , the eigenvalues  $\lambda$  of its correlation matrix follow:

$$f(\lambda) = \frac{q}{2\pi\sigma^2\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \quad \text{for } \lambda \in [\lambda_-, \lambda_+]$$

where  $\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{\frac{1}{q}}\right)^2$ . Eigenvalues within  $[\lambda_-, \lambda_+]$  represent pure noise.

**Denoising via Constant Residual Eigenvalues:** Keep eigenvalues  $\lambda > \lambda_+$  (Signal). Replace all noise eigenvalues  $\lambda \leq \lambda_+$  with their average, keeping total variance (trace) constant.

## Topic 4.4: Random Matrix Theory (Detoning)

### The "Market Component"

The largest eigenvalue  $\lambda_{\max}$  of a financial correlation matrix is typically massive, representing the market-wide dominant factor (all assets moving together).

- **The Detoning Process:**

- Financial clustering (unsupervised classification) can be blinded by this dominant market-wide component.
- Detoning removes the first eigenvector (associated with  $\lambda_{\max}$ ) from the correlation matrix:

$$C_{\text{detoned}} = C - \lambda_1 W_1 W_1^T \quad (\text{where } W_1 \text{ is the first eigenvector}).$$

- **Result:** Allows algorithms to focus on sector-specific, micro-alpha relationships and subtler corporate clusters without market-wide bias.

# Summary: The Complete Financial ML Pipeline

## The End-to-End Workflow

Raw Ticks  $\longrightarrow$  **Dollar Bars** (Ch.2)  $\longrightarrow$  **Triple-Barrier** (Ch.3)  $\longrightarrow$  **Meta-Labeling** (Ch.3)  
 $\longrightarrow$  **Denoised Covariance** (Ch.4)

## Main Takeaway

Successful financial ML is not about finding a complex algorithm. It is about building a mathematically rigorous data and labeling pipeline that respects market microstructure, path-dependency, and statistical boundaries.

# Q&A

# Thank You! Questions & Discussions

AI Finance Seminar, Sogang University